Paralleling Device(7VE85) SIPROTEC 5

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Overview of Functions

1. **Provides 2-channel redundant synchronization** for enhanced safety and reliability used for:
   I. Synchronizing a generator with the power system.
   II. Synchronizing two power systems.
2. **Independent measurement and processing** in both channels ensure reliable operation.
3. **Supports the following operating modes:**
   I. Synchronous switching
   II. Asynchronous switching
   III. Switching to a dead line or dead busbar
   IV. Direct close command
   V. Automatic voltage balancing
   VI. Automatic frequency balancing

General Functionality

The Paralleling function consists of the General and Inv. Voltage function blocks.

General Function Block:

Processes V_{sync1} and V_{sync2} to calculate the synchronization measurements for Channel 1.

Inverse Voltage Function Block:

Processes $V_{sync1, inv}$ and $V_{sync2, inv}$ to calculate the synchronization measurements for Channel 2.

Two-Channel Measurement:

Both function blocks provide independent measurements for the paralleling stages, ensuring redundant and reliable operation.

General Functionality

Reference Values:

Voltage (V_1), frequency (f_1), and phase angle (α_1) are measured from Side 1.

Measured Values:

Voltage (V_2), frequency (f_2), and phase angle (α_2) are measured from Side 2.

Three-Phase Measurement:

If both sides use three-phase measurement points, V_1 and V_2 represent the phase to phase voltage (V_{AB}).

Mixed Measurement:

If one side uses three phase measurement and the other uses single phase measurement, V_1 and V_2 represent the single-phase voltage of each side.

Single-Phase Measurement:

If both sides use single-phase measurement points, the device considers the phase angle and voltage magnitude correction factors during synchronization calculations.

General Functionality

Voltage Difference ($dV=V_2-V_1$):

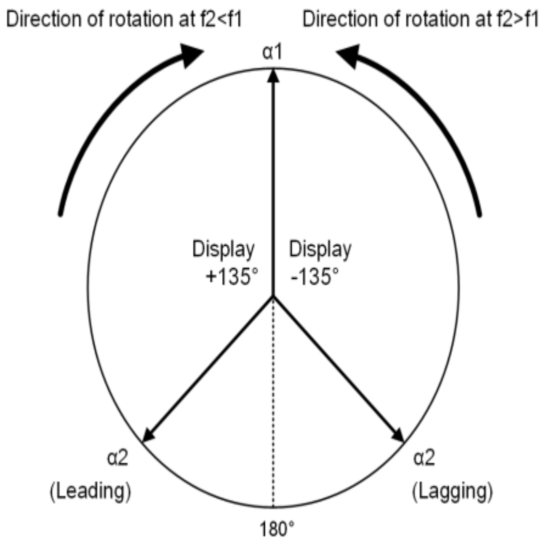
A positive value indicates $V_2 > V_1$, while a negative value indicates $V_2<V_1$.

Frequency Difference ($df=f_2-f_1$):

A positive value indicates an over synchronous condition ($f2 > f_1$), while a negative value indicates an under synchronous condition ($f_2 < f_1$).

Phase Angle Difference ($d\alpha=\alpha_2-\alpha_1$):

Limited to ±180°. A positive value means α_2 leads α_1, while a negative value means α_2 lags α_1.



Angle Rotation:

During asynchronous operation, $d\alpha$ rotates counterclockwise when $f_2 > f_1$ and clockwise when $f_2 < f_1$.

Parameter Setting Convention:

Only positive parameter values are permitted. Separate parameters are used for positive and negative deviations (e.g., $V_2 > V_1$ and $V_2 < V_1$). The same convention applies to frequency difference (df) and phase angle difference ($d\alpha$), ensuring clear and consistent parameter settings.

Measure

Diverse Measurement Algorithms

The system uses two different measurement algorithms to support multichannel redundancy.

Independent Processing

Each algorithm operates independently using separate measurement procedures and dedicated memory.

Fault Tolerance

The use of diverse algorithms minimizes the risk of systematic errors causing incorrect or unintended operation.

Enhanced Reliability

Independent measurement and processing improve the safety, reliability, and availability of the synchronization function.

Paralleling Function

Paralleling Function can be performed in two ways:

1.5-channel Paralleling Function:

- System Capacity < 100 MVA.
- Provides high reliability with a 2-out-of-2 decision.
- One channel performs the synchronization decision.

2-channel Paralleling Function:

- System Capacity ≥ 100 MVA.
- Provides maximum reliability with a 2-out-of-2 decision.
- Both channels independently verify synchronization before issuing the close command.

Stage paralleling with 2 channels

2-channel Paralleling Function

Duplicated Voltage Inputs

Voltage inputs (**V1** and **V2**) are duplicated and supplied to two independent A/D converters.

Independent Measurements

Measurement Procedures 1 and 2 process the voltage signals independently.

Separate Synchronization Functions

Each measurement is evaluated by its own synchronization function.

Synchronization Verification

Each function independently verifies:

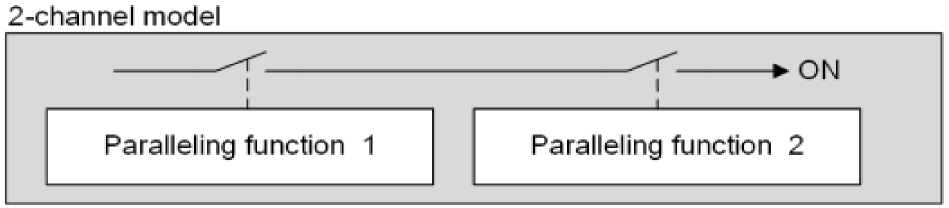
- Voltage difference
- Frequency difference
- Phase angle difference

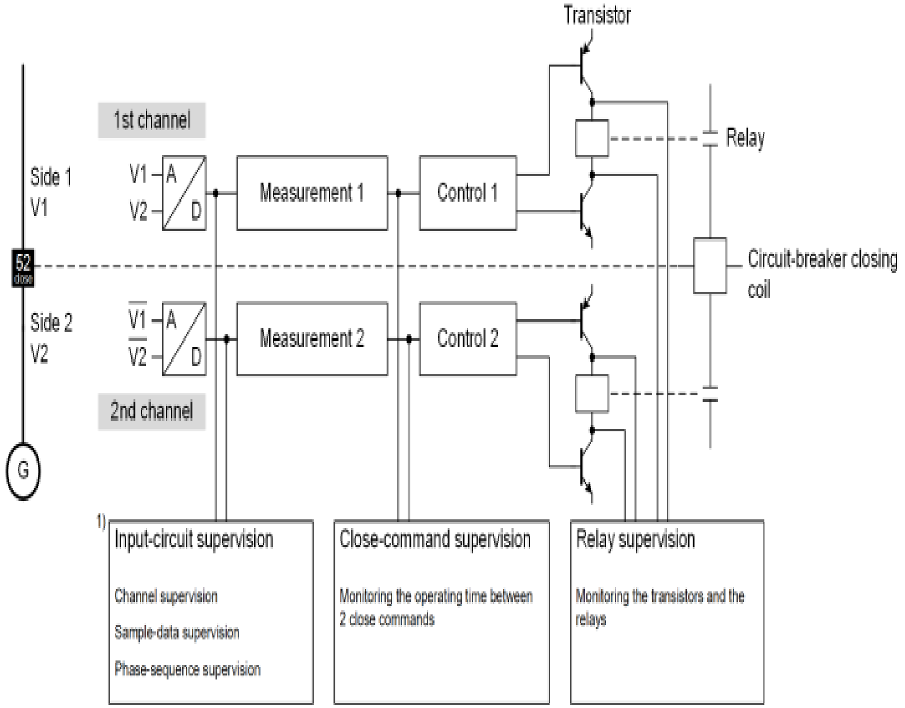
Release Decision

A **Release** signal is issued only if all synchronization criteria are satisfied.

Close Command Generation

The close command is generated only when **both synchronization functions** issue a Release.



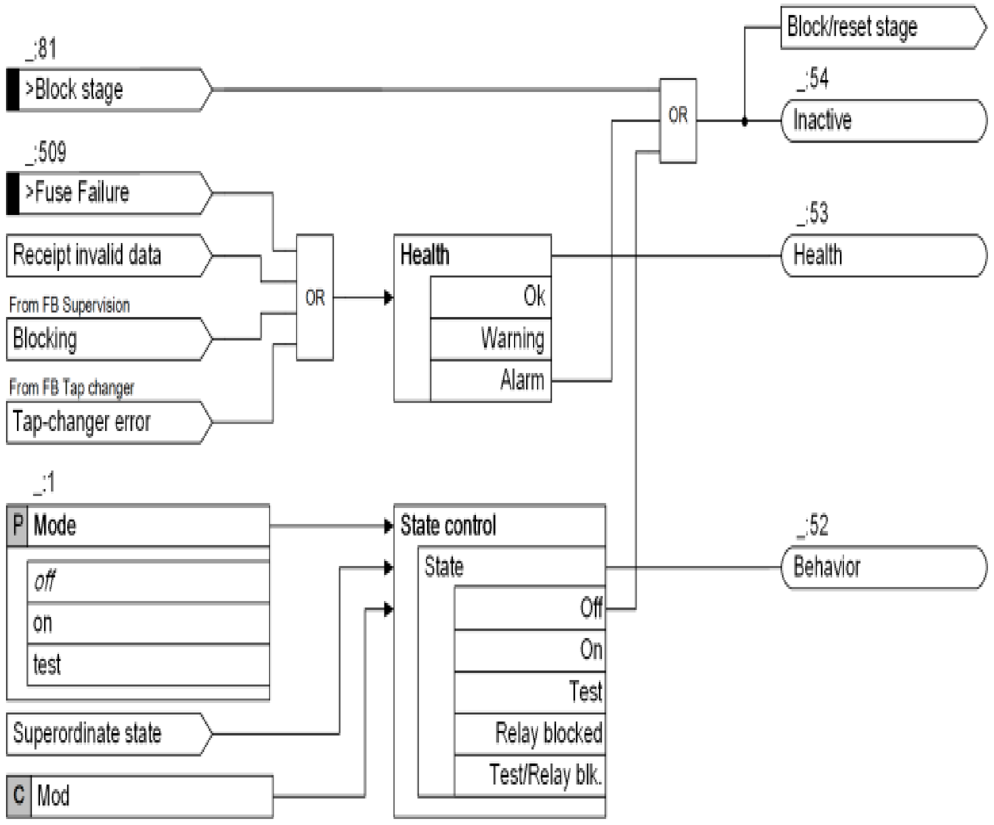


Stage paralleling with 2 channels(Logic)

Block Stage: Disables the function when an external block signal is active.
Health Monitoring: Continuously checks for abnormal conditions such as:

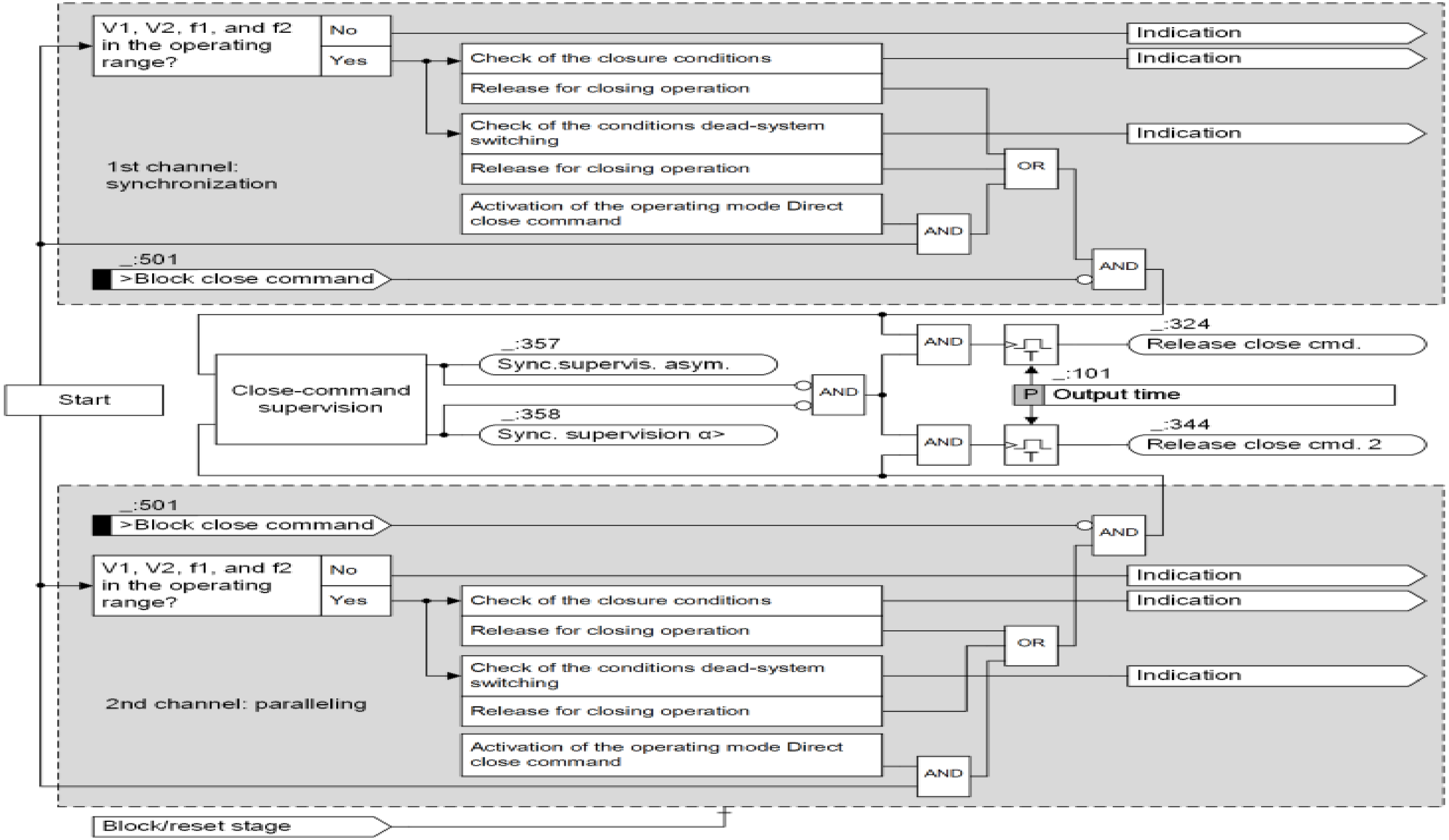
- Fuse failure
- Invalid data reception
- Blocking signal from FB Supervision
- Tap-changer error

Health Status: Reports the stage condition as OK, Warning or Alarm.
State Control: Determines the operating mode based on user settings and external commands.
Operating Modes: Supports Off, On, Test, Relay Blocked, and Test/Relay Blocked states.
Outputs: Generates Behavior, Health, and Inactive (Block/Reset) status signals for supervision and control.



Logic Diagram of the Stage Control

Stage paralleling with 2 channels

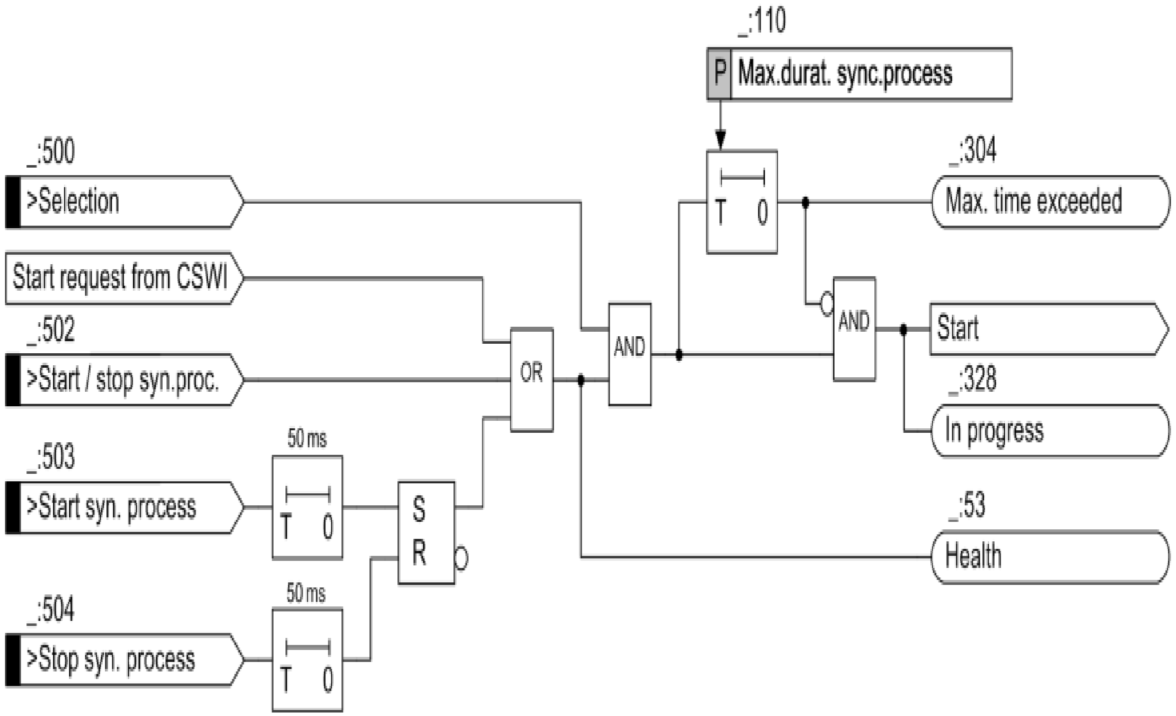


Logic Diagram of the Stage Paralleling with 2 Channels

Stage paralleling with 2 channels(Start)

The stage Paralleling with 2 channels can be started by one of the following 3 options:

1. Via the binary input signal **>Start syn. Process** which is a pulse signal. The binary input signal **>Stop syn. Process** is the corresponding stop signal of the signal **>Start. syn. Process**.
2. Via the binary input signal **>Start/stop syn.proc.** when this input signal is active.
3. Via the start request from CSWI.



4. After a successful start, the indication **In progress** is issued & the supervision time **Max.durat. sync.process** for the maximum duration of the paralleling process starts.

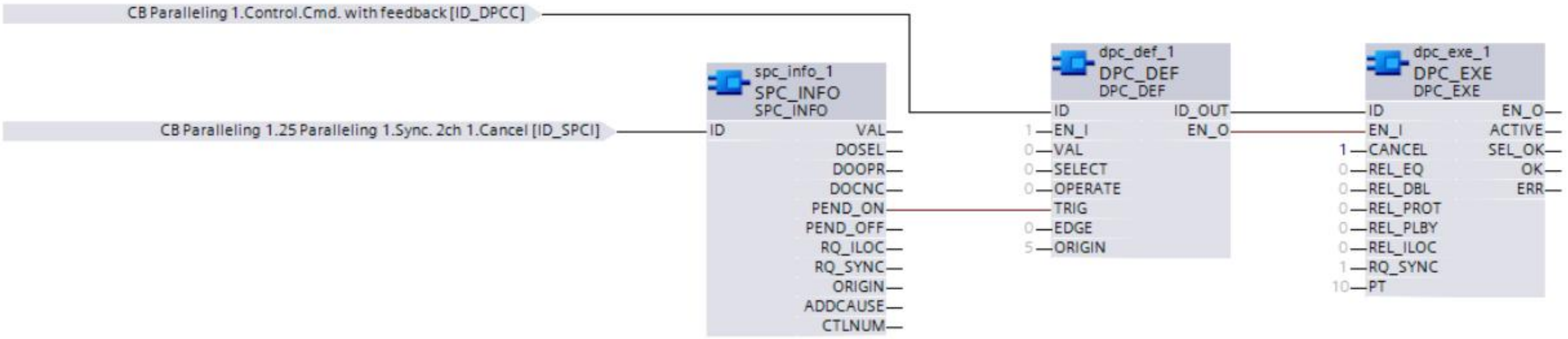
Stage paralleling with 2 channels(Stop)

The stage **Paralleling with 2 channels** can be stopped by one of the following options:

1. Via one of the following binary input signals:
   - >Stop syn. Process.
   - >Start/stop syn.proc.
   - >Fuse Failure.
2. Via the voltage transformer miniature circuit breaker:
   - If the voltage transformer miniature circuit breaker is open, the health state of the stage **Paralleling with 2 channels** changes to Warning and the stage stops.
3. Via the supervision time **Max.durat. sync.process:**
   - If the supervision time **Max.durat. sync.process** expires, the stage **Paralleling with 2 channels** stops.

Stage paralleling with 2 channels(Stop)

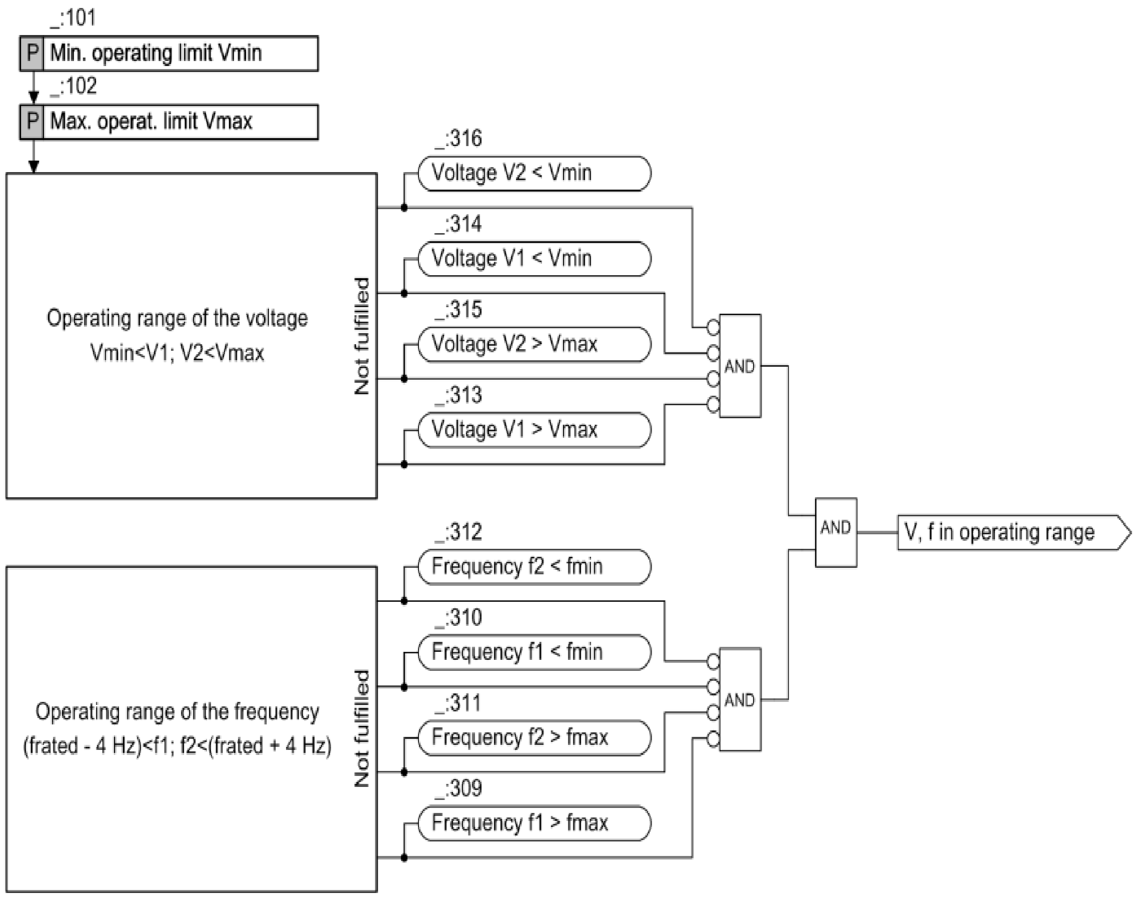
4. Via the release close commands from 2 channels:
   - When the 2 indications Release close cmd. and Release close cmd. 2 are generated simultaneously, the stage **Paralleling with 2 channels** stops.
5. Via the blocking signal from the function block **Supervision**.
6. Via CFC when the stage is started via the start request from CSWI:
   - If you start the stage via the start request from CSWI, you can stop the stage via the following
   - user-defined CFC chart:



Stage paralleling with 2 channels(Operating Range)

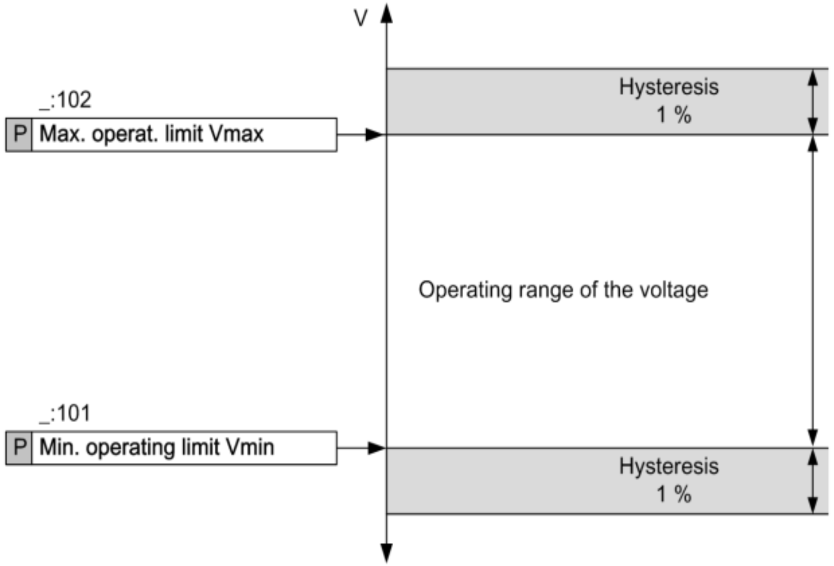
The operating range of the stage Paralleling with 2 channels is defined by the parameters **Min. operating limit V_{min}** and **Max. operat. limit Vmax &** the specified frequency band f_{rated} ± 4 Hz.

If the voltages or the frequencies are out of the permitted operating range, the corresponding indications are issued, such as **Voltage V1 < Vmin** or **Frequency f1 > fmax**.



Stage paralleling with 2 channels(Operating Range)

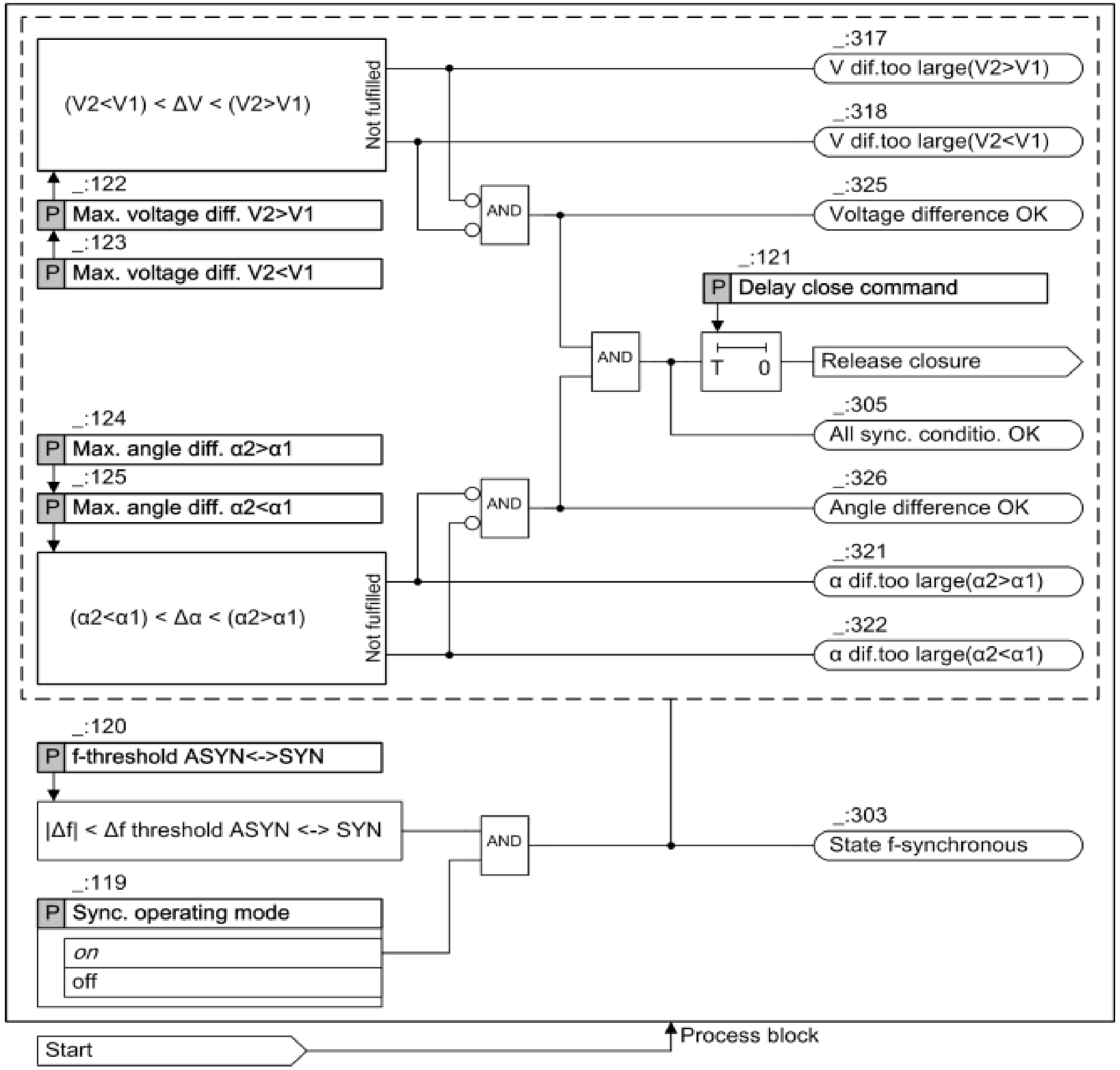
1. Configure the minimum (Vmin) and maximum (Vmax) operating voltage limits, typically **±10%** of the rated voltage.
2. The relay checks whether the voltage is within the configured operating range.
3. If the stage starts within the hysteresis area, the relay does not stop or switch because of the Vmin or Vmax limits.
4. The relay continuously monitors the voltage while the paralleling process is in progress.
5. Switching is permitted if:
   - The stage started within the operating range
   - The voltage later exceeds the minimum or maximum operating limit during the paralleling process.



Stage paralleling with 2 Channel (Checking the Closing Conditions in Synchronous Systems)

1. The frequency difference (Δf) is below the **ASYN↔SYN threshold**, and the system enters **frequency-synchronous mode**.
2. The relay checks **voltage difference (ΔV)** and **phase angle difference (Δα)** against the configured limits. If both conditions are satisfied, **All Sync. Conditions OK** is indicated.
3. After the configured **Delay Close Command** expires, the relay issues the **closure release**.
4. Individual status indications confirm each condition:
   1. **Voltage Difference OK**
   2. **Angle Difference OK**
5. If any condition is outside the limit, the relay displays a specific indication (e.g., **V diff. too large (V2 < V1)**), helping the operator identify the required correction before paralleling.

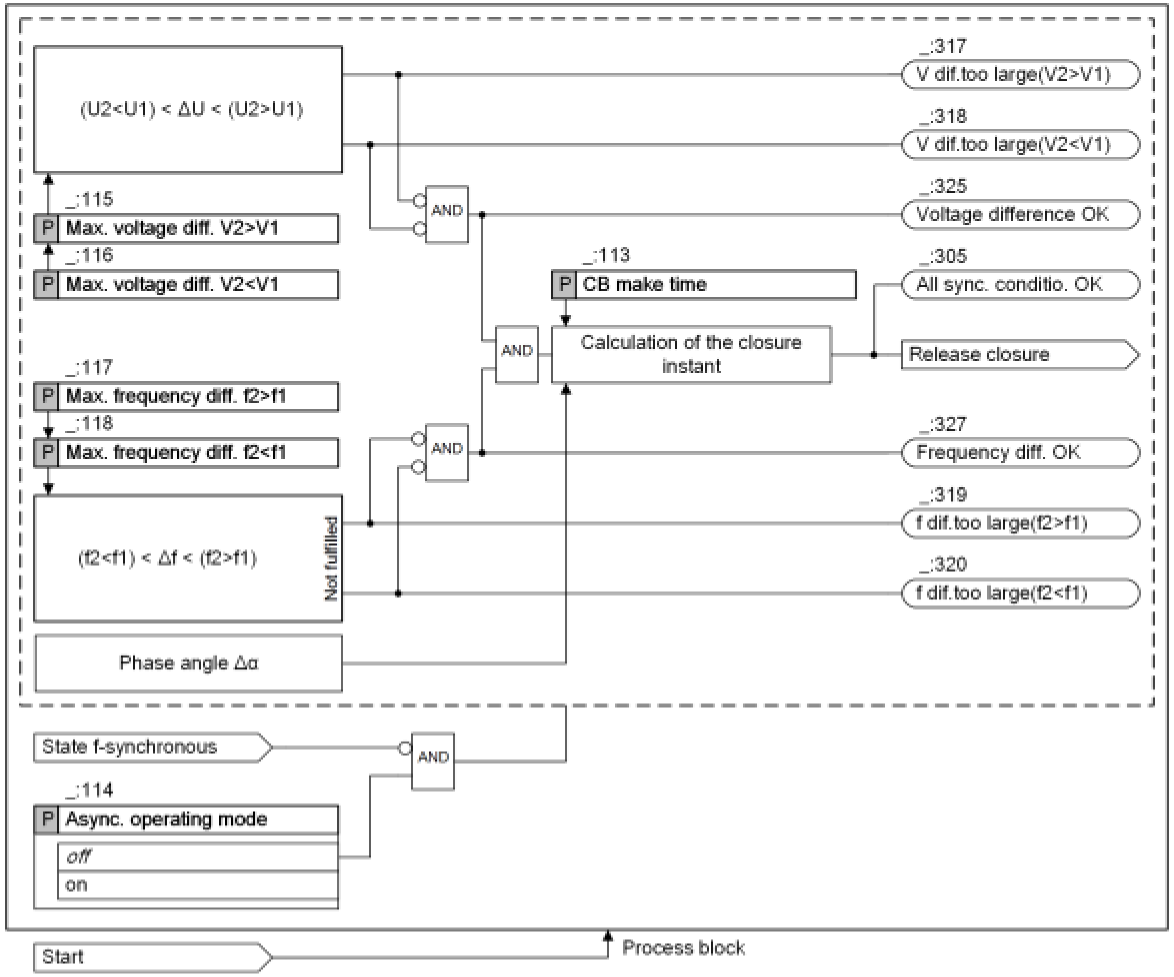
The circuit breaker is permitted to close only when the frequency, voltage, and phase angle are all within the preset synchronization limits.



Stage paralleling with 2 Channel (Checking the Closing Conditions in Asynchronous Systems)

1. The relay continuously monitors **voltage difference (ΔV)** and **frequency difference (Δf)**.
2. It calculates the **optimum close command timing** using:
   - **Phase angle difference (Δα)**
   - **Circuit breaker make time**
3. The close command is issued so that, at the instant the breaker closes:
   - **ΔV ≈ 0**
   - **Δα ≈ 0**
4. This ensures both voltage phasors are aligned for **safe and accurate synchronization**.
5. The **frequency difference (Δf)** must remain small; otherwise, rapid changes in phase and voltage make accurate synchronization unstable and unreliable.

Result: The relay predicts the ideal closing instant to achieve synchronization at the moment of circuit breaker closure.



Stage paralleling with 2 Channel (Switching to Dead Line or Dead Busbar)

Verify Dead Busbar:

Confirm the busbar voltage (V2) is below the **"without voltage"** threshold.

Verify Live Source:

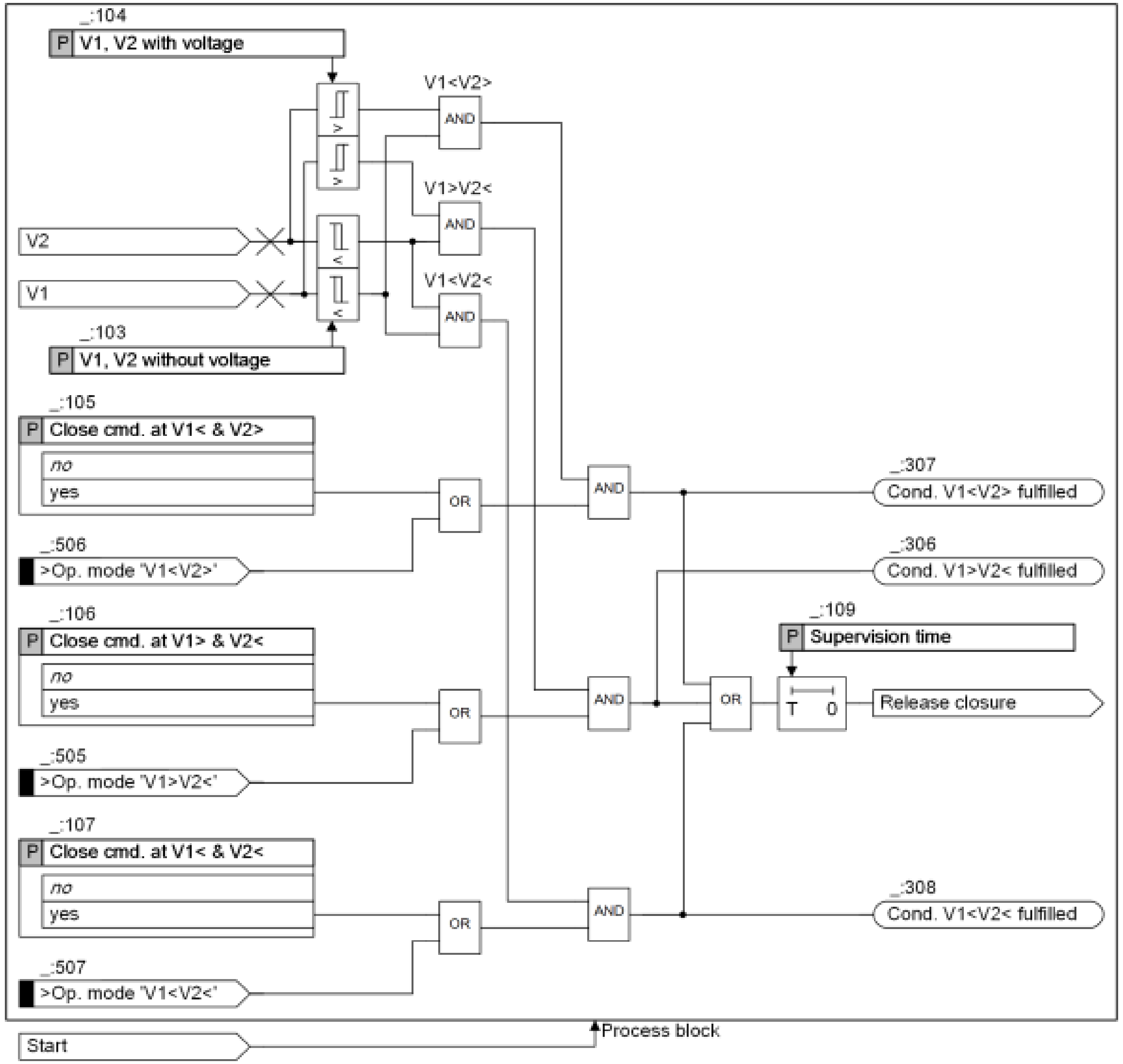
Confirm the source voltage (V1) is above the **"with voltage"** threshold and within the permissible voltage and frequency limits.

Start Supervision Time:

The relay monitors that the dead/live conditions remain stable for the configured supervision time.

Issue Close Release:

If all conditions remain valid, the relay issues the closing permission and the circuit breaker closes to energize the dead busbar.



Stage paralleling with 2 Channel (Direct Close Command)

Enable the function either:

- Statically using the Direct close command parameter, or
- Dynamically using the binary input >Op. mode 'dir.cls.cmd'.

Start the Paralleling Stage:

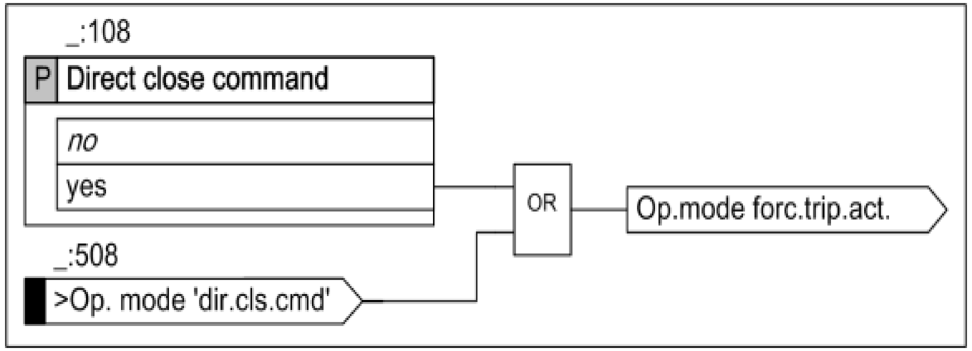
The Paralleling with 2 channels stage is initiated.

No Condition Checks:

The relay does not perform any synchronization, voltage, frequency, phase angle, or dead-system checks.

Immediate Close Release:

The relay immediately generates the Close Release signal upon stage start.



Stage paralleling with 2 Channel(Close-Command Monitoring**)**

Monitor Both Channels:

The relay monitors the close commands generated by Channel 1 and Channel 2.

Check Condition Consistency:

The relay verifies that both channels are operating under the same condition (e.g., both synchronized or both dead-bus).

If the conditions differ, the close command is blocked and Sync.supervis. asym. is indicated.

1-Minute Mismatch Timer:

If the condition mismatch continues for 1 minute, the Paralleling function stops and all signals are reset.

Check Simultaneous Close Commands:

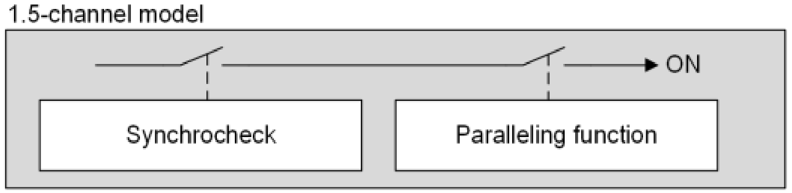
The relay checks that both channels generate their close commands at the same time.

Block Closing if Commands Differ:

If the two close commands are not generated simultaneously, the close command is blocked and the indication Sync. supervision α> is issued.

Stage paralleling with 1.5 Channel

1.5-channel Paralleling Function



Duplicate voltage inputs (V1 & V2):

- Voltages are measured independently by two separate procedures.

Measurement Procedure 1(Synchrocheck):

- Verifies voltage, frequency, and phase angle against preset limits.
- Provides the release (permission) signal for synchronization.

Measurement Procedure 2(Synchronization):

- Calculates the optimum circuit breaker closing instant.
- Issues the close command only after receiving the Synchrocheck release.

Circuit Breaker Operation:

- The breaker closes only when both functions are active and all synchronization conditions are satisfied.

Logic of Stage Control

Block Stage: Disables the function when an external block signal is active.
Health Monitoring: Continuously checks for abnormal conditions such as:

- Fuse failure
- Invalid data reception
- Blocking signal from FB Supervision
- Tap-changer error

Health Status: Reports the stage condition as OK, Warning or Alarm.
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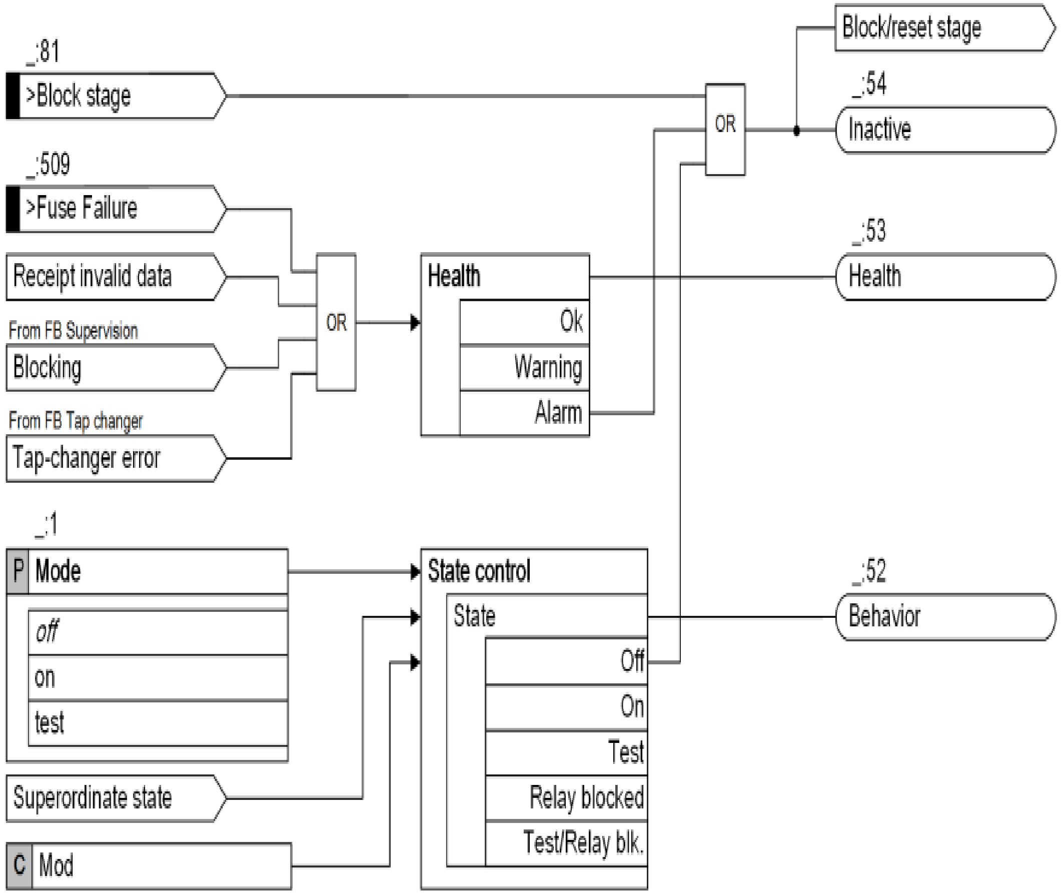


Logic Diagram of the Stage

Channel 1 (0.5 Channel) Synchrocheck:

- Verifies synchronization criteria.
- Provides the **release signal** for the close command.

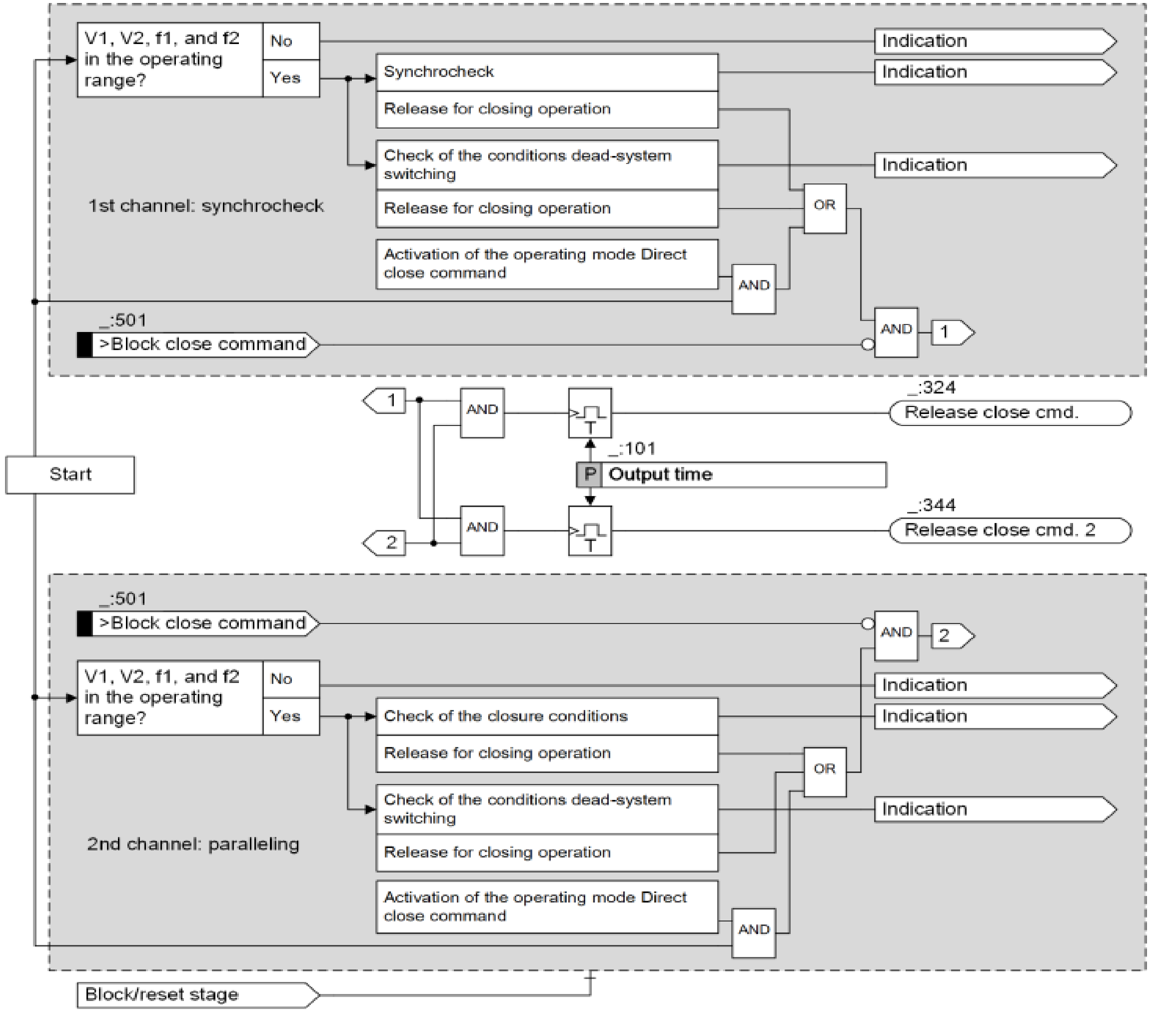
Channel 2 Synchronization:

- Automatically detects the optimum closing condition.
- Supports synchronization under different closing conditions.

Key Functionalities:

- Start and stop control
- Configurable operating range
- Closing condition verification in synchronous systems
- Closing condition verification in asynchronous systems

These functions are same as describes in Paralleling with 2 channels.



Closing Conditions in the Synchrocheck

Before circuit breaker closing, the relay independently checks:

- Voltage difference (ΔV)
- Frequency difference (Δf)
- Phase angle difference ($\Delta\alpha$)

All Sync. Conditions OK:

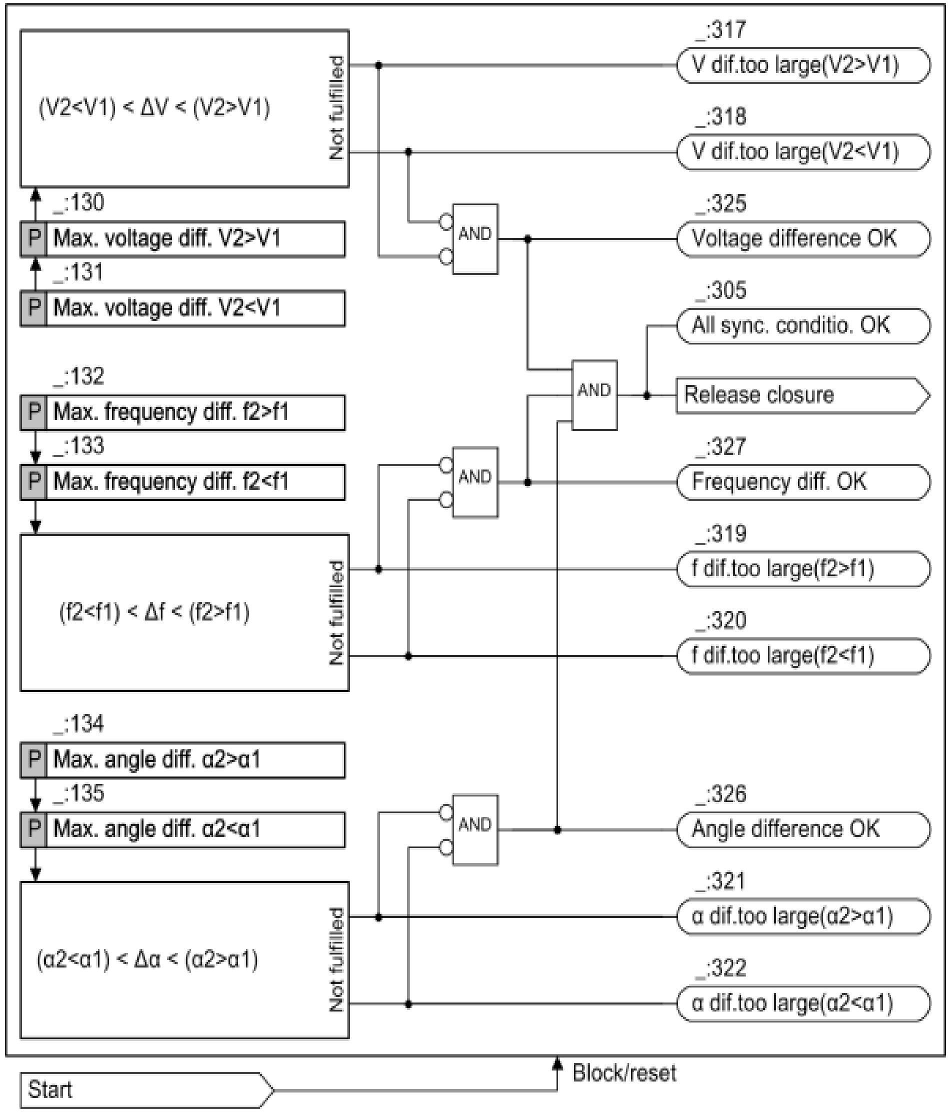
Indicates that all preset synchronization limits are satisfied.

Each parameter is monitored individually and confirmed by:

- Voltage Difference OK
- Frequency Difference OK
- Angle Difference OK

If any condition is outside the allowable limits, a specific indication identifies the cause.

Example: "V diff. too large (V2 < V1)" indicates the voltage difference exceeds the limit and V2 must be increased before successful paralleling is possible.



Stage Balancing Commands

Stage Balancing Commands

1. **Works with Paralleling:** The Balancing commands stage operates only with Paralleling with 1.5 channels or Paralleling with 2 channels. It cannot operate independently.
2. **Purpose:** It provides automatic voltage and frequency balancing to synchronize generators before closing.
3. **Voltage Inputs:** The function uses the 2nd channel voltage inputs, $V_{sync1,\ inv}$ and $V_{sync2,\ inv}$, for balancing calculations.
4. **Supports Automatic Generator Synchronization:** The relay issues balancing commands to adjust the generator until synchronization conditions are achieved.
5. **DIGSI 5 Location:** In DIGSI 5, the stage is located under: Global Library→FG Circuit Breaker→Paralleling Function→Function Extensions→Balancing Commands.

Stage Balancing Commands for Voltage & Frequnecy

The stage has 2 separate blocks:

1. Block 1: Voltage Balancing.
2. Block 2: Frequency Balancing.

The following values affect the pulse width of the balancing command:

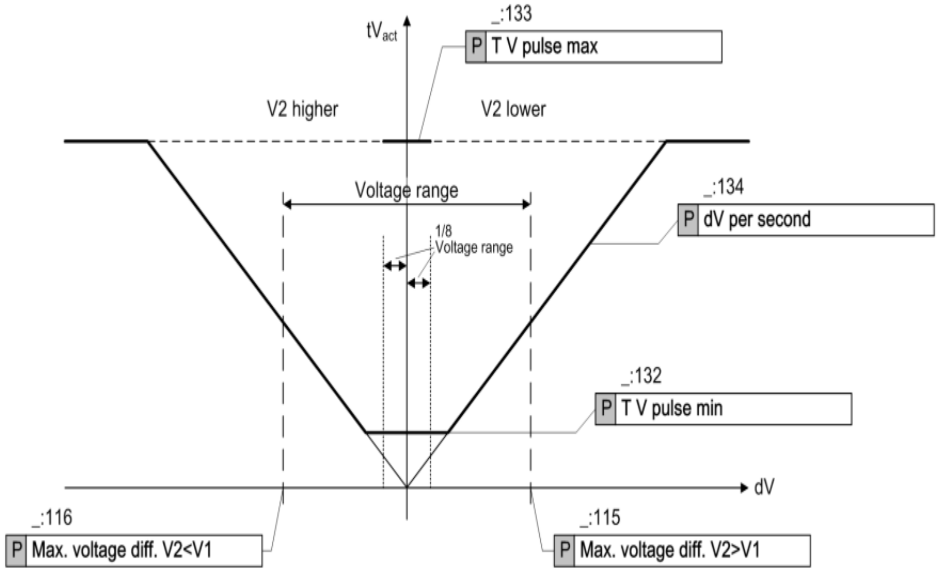
1. The current measured values of the differential voltage.
2. The current measured values of the differential frequency.
3. The setting values of dV/dt and df/dt.

In addition, the minimum pulse duration (**TV pulse min**, **Tf pulse min**) and the maximum pulse duration (**TV pulse max**, **Tf pulse max**) are defined. The minimum pulse duration allows a safe reaction of the controllers while the maximum pulse duration prevents an overreaction.

The balancing commands are active once the **Paralleling** function is started.

Stage Balancing Commands for Voltage

- The parameters **Max. voltage diff. $V_2>V_1$** and **Max. voltage diff. $V_2<V_1$** are used to determine the voltage range that defines the admissible voltage difference for asynchronous parallel switching. The median value of the voltage range is defined as the set point value for the balancing command.
- The parameters **TV pulse min** and **TV pulse max** define the minimum and maximum duration of the balancing pulse.
- The parameter **dV per second** defines the voltage-changing speed of the side to be paralleled. The functionality uses the measured voltage difference and the setting value of the parameter **dV per second** to determine the actuating time. The following equation applies.
- The following figure shows the effect of the functionality. When the voltage range falls below approximately 1/8 of the voltage range, the stage no longer generates any voltage balancing command.



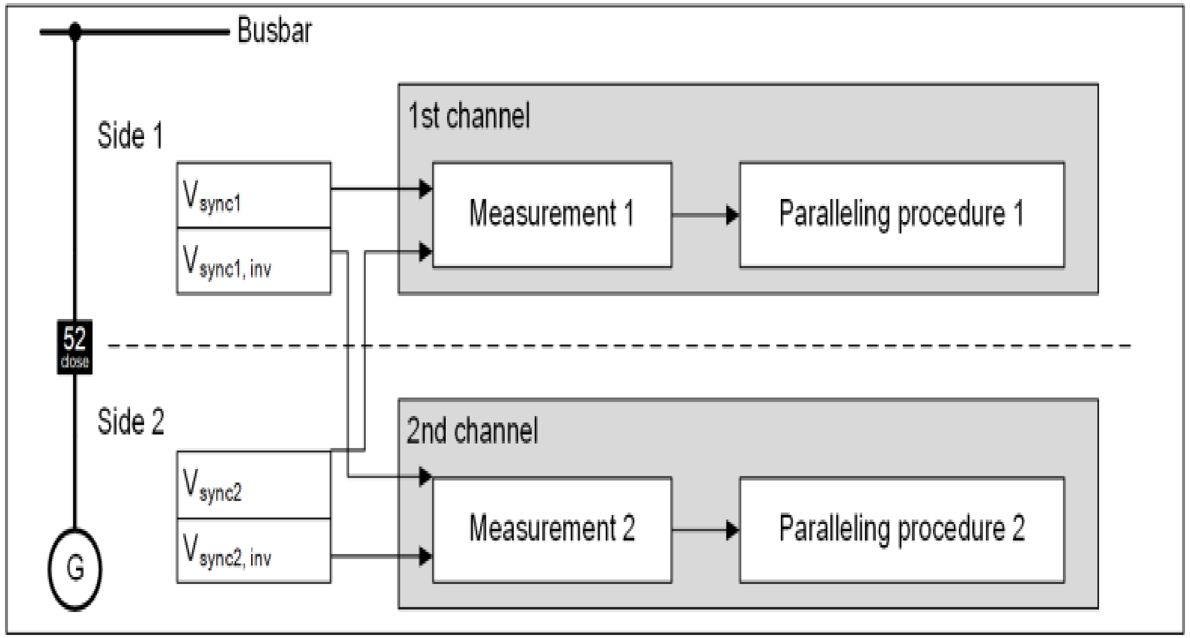
$$\mathrm{tV_{act}} = \frac{dV_{MV}}{dV/dt}$$

$\mathrm{tV_{act}}$	Pulse duration
dV_{MV}	Measured voltage difference
dV/dt	The setting value of the parameter **dV per second** for defining the voltage-changing speed of the side to be paralleled

VT Connections

Four standard voltage connection configurations are available for the Paralleling function.

- Side 1 (Reference Side) is typically connected to the busbar. Provides $\mathbf{V_{sync1}}$ and $\mathbf{V_{sync1,\ inv}}$.
- Side 2 (Incoming Side) is typically connected to the generator or incoming source. Provides $\mathbf{V_{sync2}}$ and $\mathbf{V_{sync2,\ inv}}$.



- $V_{sync,\ inv}$ is the inverse (180° opposite polarity) of the corresponding $V_{sync\ voltage}$.
- V_{sync} and $V_{sync,\ inv}$ are assigned to different synchronization channels, enabling independent synchronization processing and improving system reliability.

VT Connections 1

Two-Phase VT Connection:

Uses two-phase isolated voltage transformers on both sides for synchronization.

Phase-Rotation Supervision:

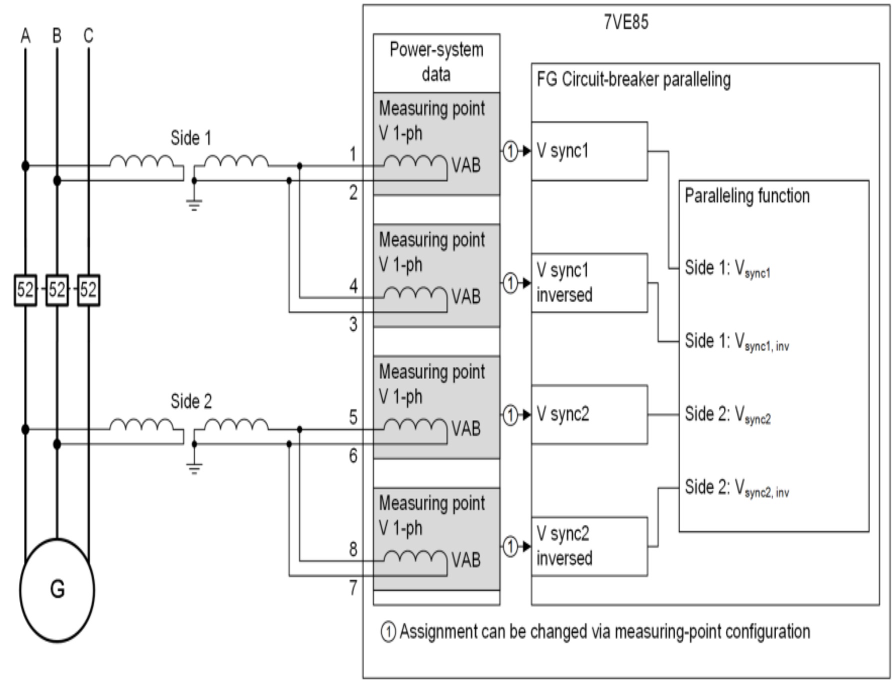
Phase-rotation supervision is unavailable because only two-phase voltages are measured.

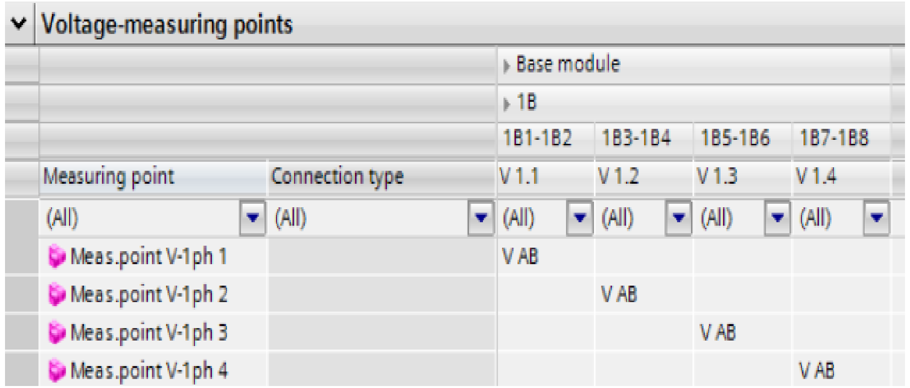
Dead Busbar Operation:

Extra care is required during dead busbar synchronization due to the lack of phase-rotation supervision.

Two-Channel Redundancy:

Provides full two-channel redundant operation for reliable and safe synchronization.





Voltage-measuring points

		Base module			
		1B			
		1B1-1B2	1B3-1B4	1B5-1B6	1B7-1B8
Measuring point	Connection type	V 1.1	V 1.2	V 1.3	V 1.4
(All)	(All)	(All)	(All)	(All)	(All)
Meas.point V-1ph 1		V AB			
Meas.point V-1ph 2			V AB		
Meas.point V-1ph 3				V AB	
Meas.point V-1ph 4					V AB

VT Connections 2

Recommended Connection:

Siemens recommends using three star-connected voltage transformers (VTs) as the standard connection.

Higher Reliability:

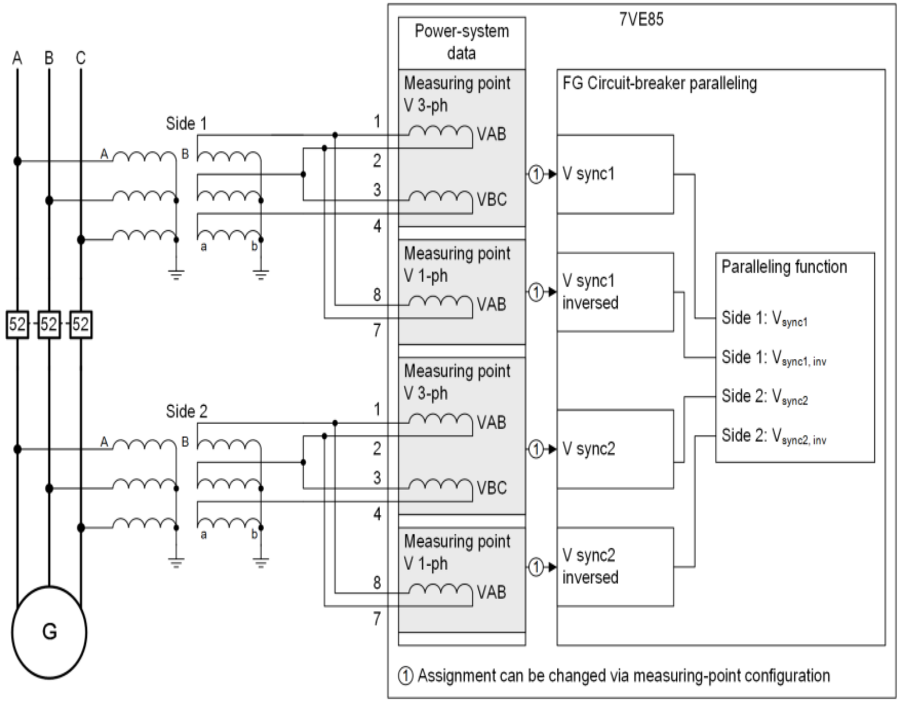
This configuration provides greater reliability for the Paralleling function than other connection methods.

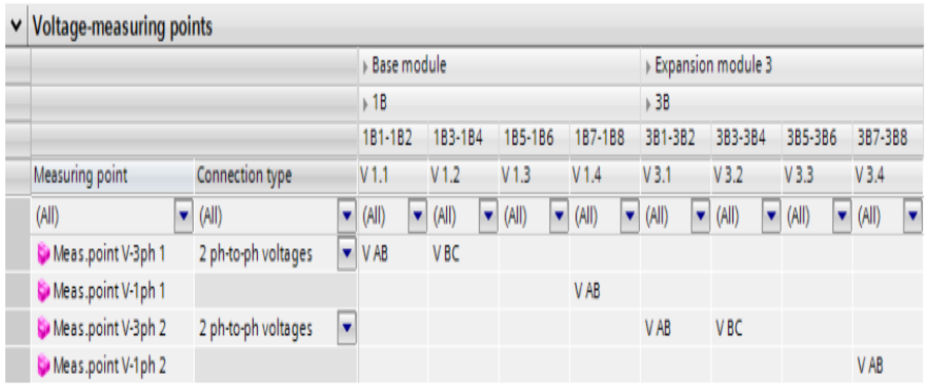
Phase Rotation Supervision:

The Voltage Phase-Rotation Supervision function is active, enabling phase-sequence verification.

Improved Safety:

Voltage connection errors are detected and do not result in unintended synchronization or circuit breaker operation.





Voltage-measuring points

		Base module				Expansion module 3			
		1B				3B			
		1B1-1B2	1B3-1B4	1B5-1B6	1B7-1B8	3B1-3B2	3B3-3B4	3B5-3B6	3B7-3B8
Measuring point	Connection type	V 1.1	V 1.2	V 1.3	V 1.4	V 3.1	V 3.2	V 3.3	V 3.4
(All)	(All)	(All)	(All)	(All)	(All)	(All)	(All)	(All)	(All)
Meas.point V-3ph 1	2 ph-to-ph voltages	V AB	V BC						
Meas.point V-1ph 1					V AB				
Meas.point V-3ph 2	2 ph-to-ph voltages					V AB	V BC		
Meas.point V-1ph 2									V AB

VT Connections 3

Recommended Connection:

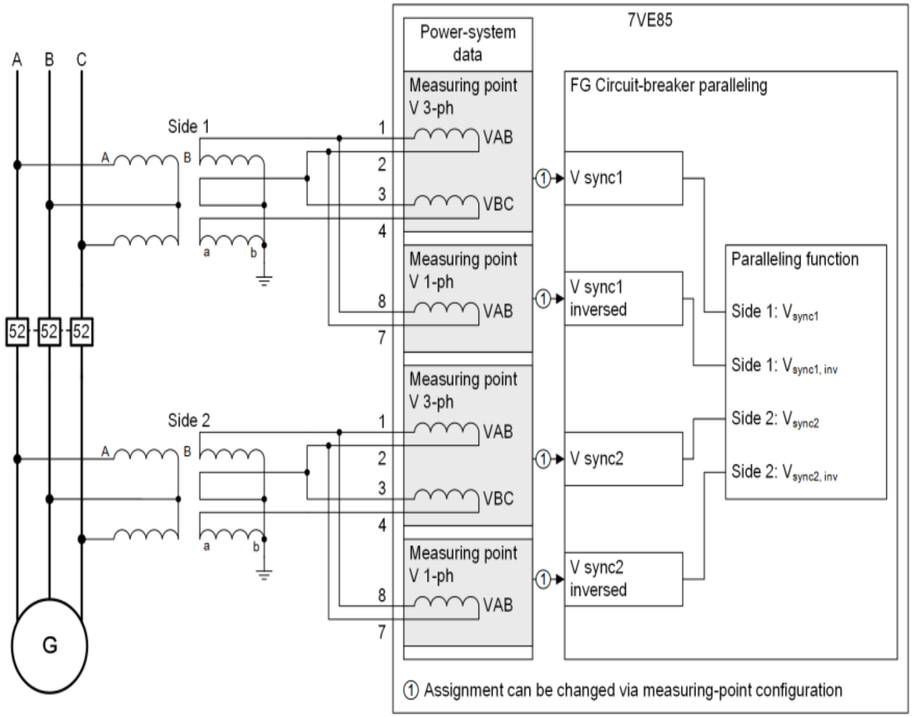
Siemens recommends using V-connected voltage transformers (VTs) where available.

Equivalent Performance:

For the Paralleling function, this connection provides electrical performance equivalent to that of three star-connected VTs.

Flexible Configuration:

A three-phase VT can be connected on one side and a V-connected VT on the other side without affecting the paralleling function.



Voltage-measuring points

		Base module				Expansion module 3			
		1B				3B			
		1B1-1B2	1B3-1B4	1B5-1B6	1B7-1B8	3B1-3B2	3B3-3B4	3B5-3B6	3B7-3B8
Measuring point	Connection type	V 1.1	V 1.2	V 1.3	V 1.4	V 3.1	V 3.2	V 3.3	V 3.4
(All)	(All)	(All)	(All)	(All)	(All)	(All)	(All)	(All)	(All)
Meas.point V-3ph 1	2 ph-to-ph voltages	V AB	V BC						
Meas.point V-1ph 1					V AB				
Meas.point V-3ph 2	2 ph-to-ph voltages					V AB	V BC		
Meas.point V-1ph 2									V AB

VT Connections 4

Single-Phase VT Connection:

The **Paralleling** function can operate using single phase isolated voltage transformers (VTs).

Ground Fault Limitation:

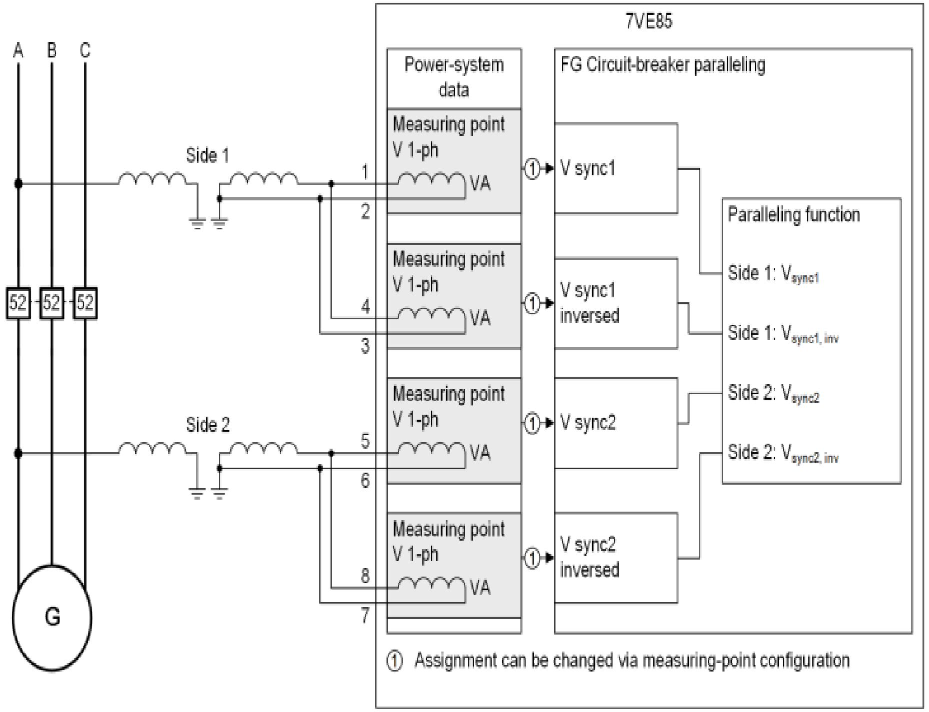
The device cannot distinguish between a phase to ground fault and a dead line or dead busbar.

Risk of Incorrect Paralleling:

If the V1< & V2> mode is enabled, a faulted busbar may be incorrectly identified as dead, increasing the risk of improper synchronization.

Phase Angle Error:

Using healthy phases during a ground fault may result in synchronization with an approximate 30° phase angle error.



Voltage-measuring points

		Base module			
		1B			
		1B1-1B2	1B3-1B4	1B5-1B6	1B7-1B8
Measuring point	Connection type	V 1.1	V 1.2	V 1.3	V 1.4
(All)	(All)	(All)	(All)	(All)	(All)
Meas.point V-1ph 1		V A			
Meas.point V-1ph 2			V A		
Meas.point V-1ph 3				V A	
Meas.point V-1ph 4					V A

Reference

SIPROTEC 5 Paralleling Device 7VE85
V11.00 and higher

Manual: C53000-G5040-C071-H